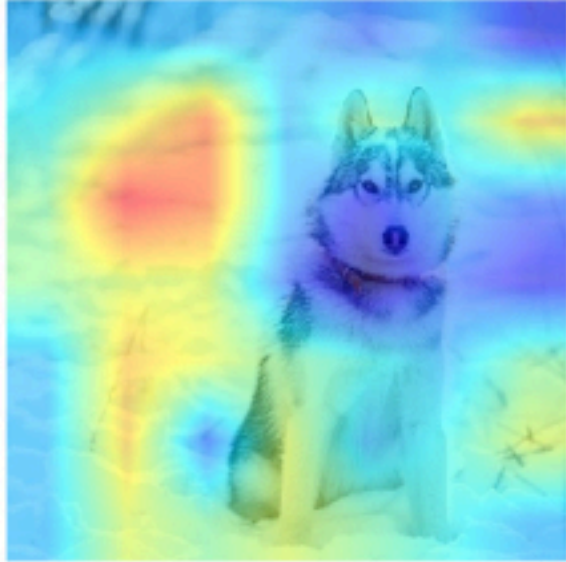
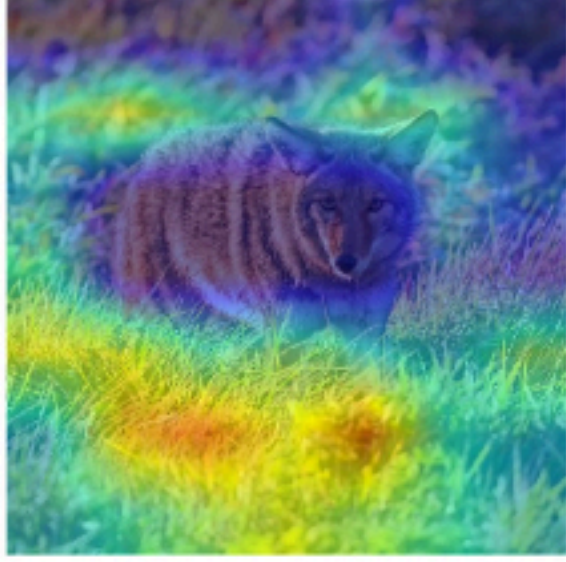


Context Attribution for RAG

VNG Internship Progress - Weeks 1-3

Eva Winston

Training sample	Test samples	Saliency map
		
Husky	Predict: Wolf	The presence of snow
		
Wolf	Predict: Husky	The presence of grass

Accuracy $\neq$
a trustworthy
model



Explainable AI (XAI)

Outline

1

WHAT IS XAI

2

PROBLEM

3

DEMO

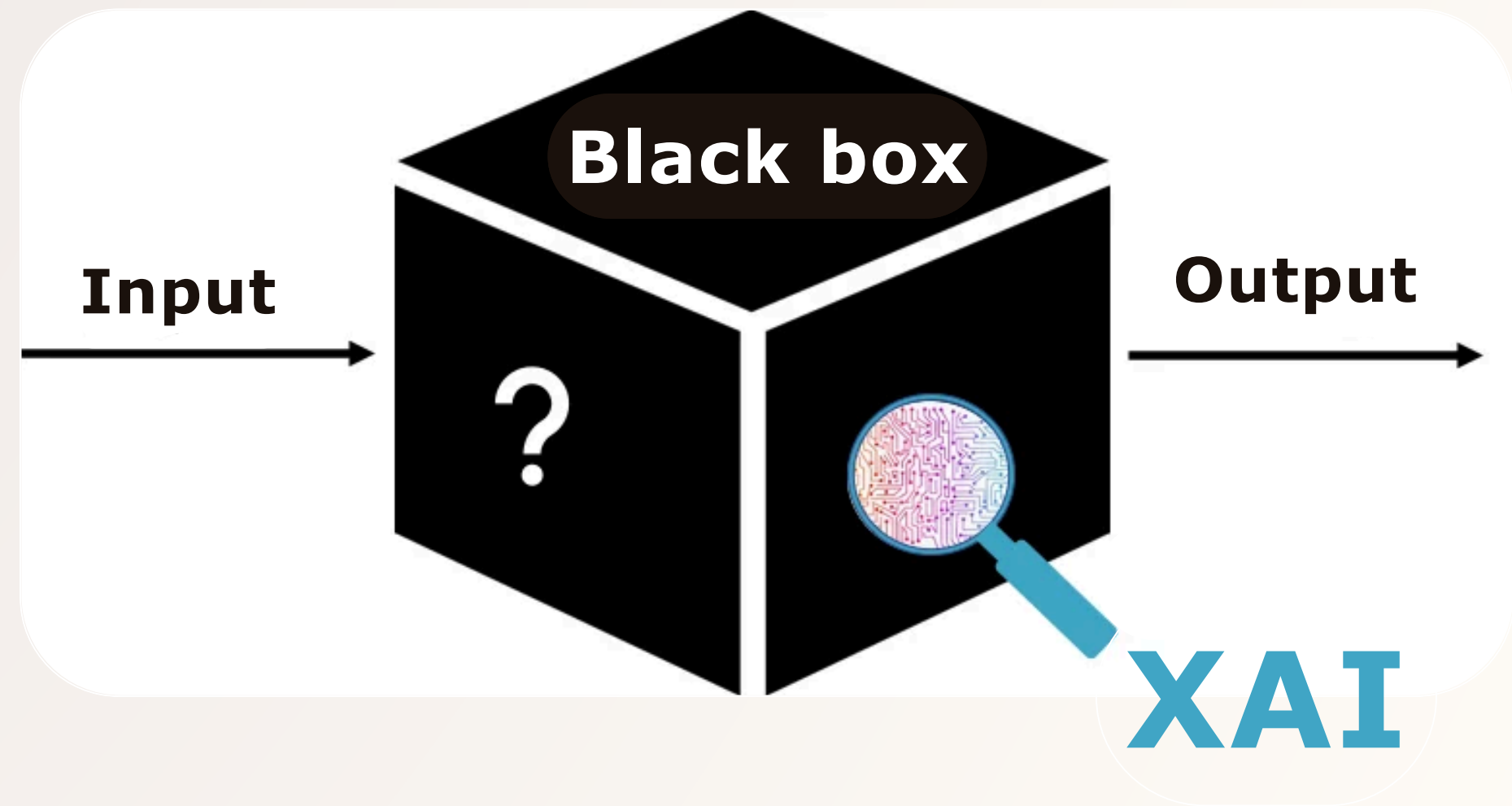
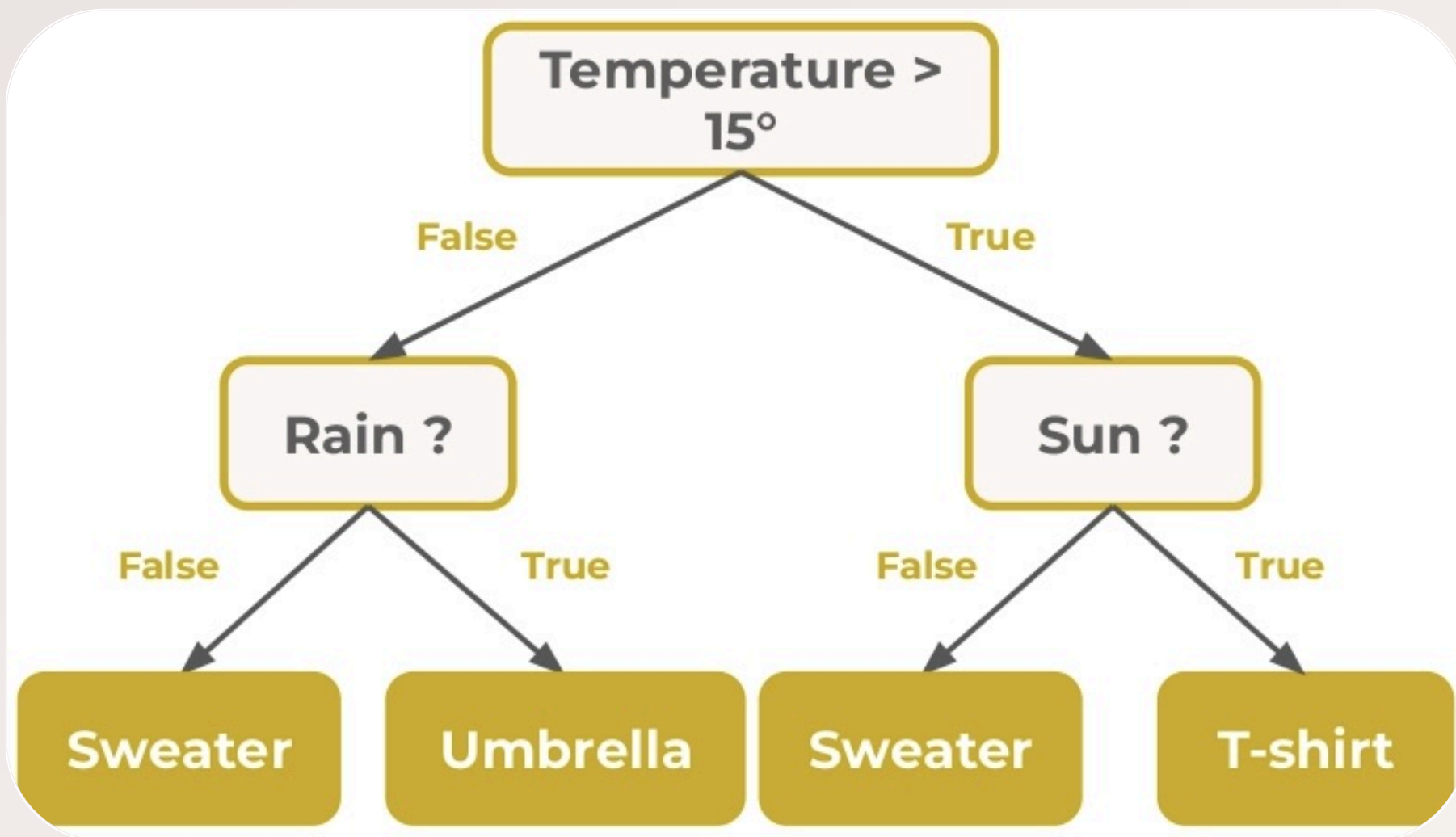
4

METHOD

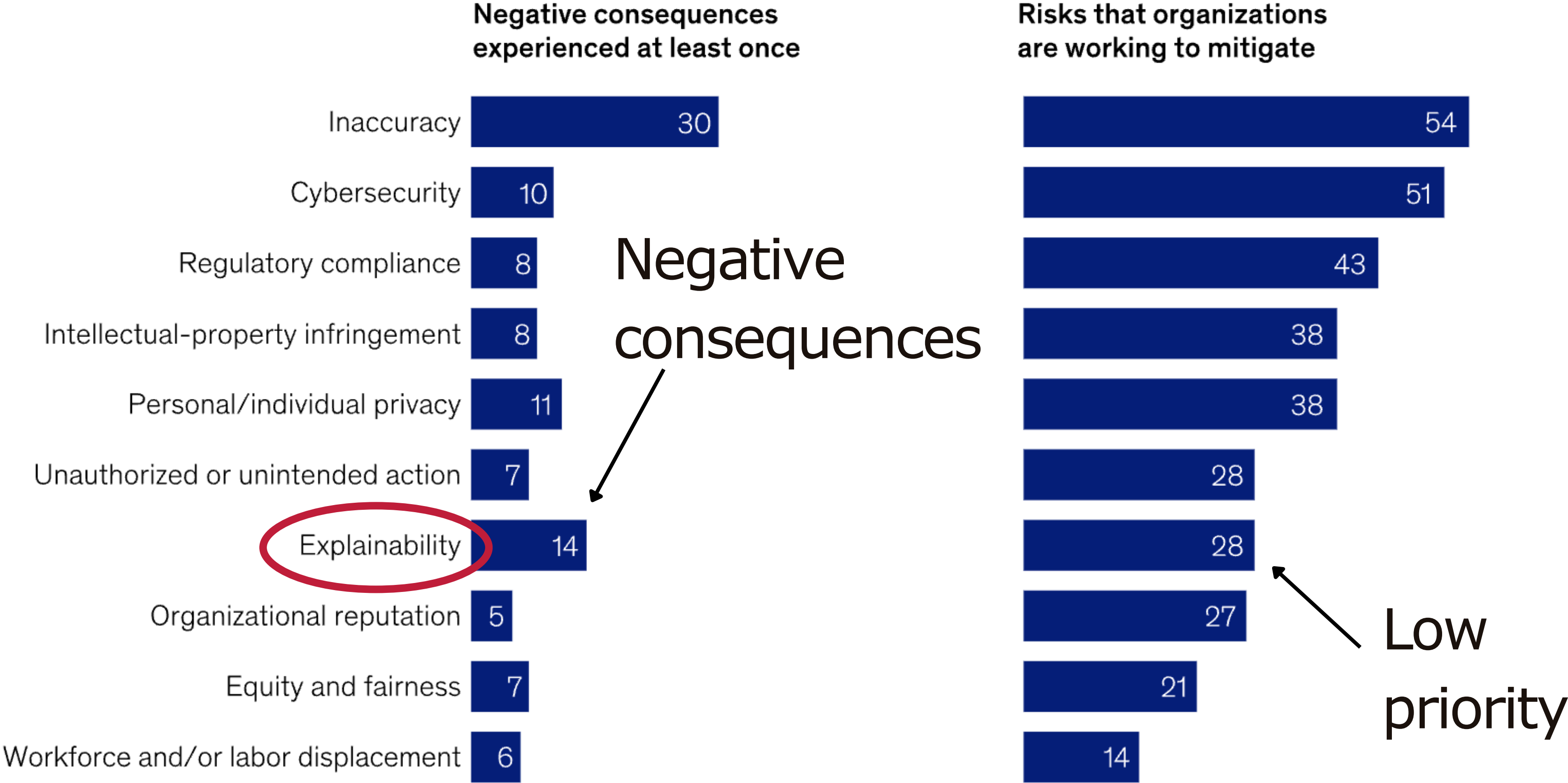
5

NEXT STEPS

XAI: Interpretability vs. Explainability



Negative consequences and risk mitigation in the past year,¹ % of respondents (n = 1,753)



Why XAI for VNG's RAG-bot?

Reasons for early implementation of context attribution

REGULATION

XAI laws are inevitable

DOMAIN

Policy enters legal territory

SCALE

Get ahead of the curve

RAG

Natural structure for attribution

What XAI attribution should do in the pipeline:

- Help explain RAG behavior
- Identify what context *caused* the response
- Build trust in AI systems internally and for stakeholders

- Detect and diagnose failures
- Allow developers to see where errors are localized
- Support efficient debugging

Demo

Example of CONTEXTCITE attributions for Llama-3-8B on CNN DailyMail

Query

Please summarize the article in up to three sentences.

Response (attributed statement is highlighted)

Here is a summary of the article:

Paul Bell, a director of online gambling firm 666Bet, was arrested at Heathrow airport as part of a £21m fraud probe, along with six others. The firm, which has deals with West Bromwich Albion and Leicester City, had signed Harry Redknapp to appear in its adverts, but Redknapp claims he knows nothing about the investigation. The firm’s licence was suspended by the UK Gambling Commission, and its website is currently offline, with no indication of when it will be restored.

CONTEXTCITE attribution

Score	Source (from the context)
34.3	... A director of an online gambling firm linked to Harry Redknapp and two Premier League football clubs has been arrested over a £21m fraud probe, it has emerged. Paul Bell, a director of 666Bet, was reportedly arrested at Heathrow airport last week as part of a joint money-laundering investigation by HM Revenue & Customs and the National Crime Agency. Six other people were arrested, 13 properties were raided and £1m in cash was also seized as part of the investigation. ...
19.1	... A director of an online gambling firm linked to Harry Redknapp and two Premier League football clubs has been arrested over a £21m fraud probe, it has emerged. Paul Bell, a director of 666Bet, was reportedly arrested at Heathrow airport last week as part of a joint money-laundering investigation by HM Revenue & Customs and the National Crime Agency. ...
18.4	... Paul Bell, a director of 666Bet, was reportedly arrested at Heathrow airport last week as part of a joint money-laundering investigation by HM Revenue & Customs and the National Crime Agency. Six other people were arrested, 13 properties were raided and £1m in cash was also seized as part of the investigation. A director of an online gambling firm, which signed Harry Redknapp to star in its adverts (pictured above) has been arrested over a £21m fraud
10.8	... Six other people were arrested, 13 properties were raided and £1m in cash was also seized as part of the investigation. A director of an online gambling firm, which signed Harry Redknapp to star in its adverts (pictured above) has been arrested over a £21m fraud . The online firm announced deals with West Bromwich Albion and Leicester City last summer. ...

Query

Please summarize the article in up to three sentences.

Attributing chars [0, 141):

Paul Bell, a director of online gambling firm 666Bet, was arrested at Heathrow ai

Paul Bell, a director of online gambling firm 666Bet, was arrested at Heathrow airport as part of a £21m fraud probe, along with six others. The firm, which has deals with West Bromwich Albion and Leicester City, had signed Harry Redknapp to appear in its adverts, but Redknapp claims he knows nothing about the investigation. The firm’s licence was suspended by the UK Gambling Commission, and its website is currently offline, with no indication of when it will be restored.

Context attribution

Score	Source
33.629	Paul Bell, a director of 666Bet, was reportedly arrested at h airport last week as part of a joint money-laundering invest: by HM Revenue & Customs and the National Crime Agency.
13.820	A director of an online gambling firm linked to Harry Redknap two Premier League football clubs has been arrested over a £2 probe, it has emerged.
8.792	Six other people were arrested, 13 properties were raided and cash was also seized as part of the investigation.
5.222	A director of an online gambling firm, which signed Harry Red star in its adverts (pictured above) has been arrested over a fraud .

Context Attribution

Upload test file (.json)

test_hallucination.json

4.4 KB ↓

Query

What is the policy regarding taxi allowances for employees?

Run

According to the company's travel guidelines, employees are entitled to a standard taxi allowance of \$50 per day for commuting between the office and their residence. This allowance is fully covered if requested via the employee portal under the 'Ground Transportation' tab, and any excess amounts up to \$100 will be automatically reimbursed if a valid receipt is provided within 5 business days.

Select a span to attribute

Full endpoint output

Context attribution

Score	Source
1.942	* **Corporate Hosting:** If an employee hosts a business dinner for clients, expenses must be filed separately under the Corporate Entertainment Policy rather than using individual per diem lines.
0.000	* **Partial Travel Days:** On the first and last days of travel, the per diem allocation is reduced to 75% of the standard daily rate (\$56.25).
-0.000	* **Daily Cap:** A fixed per diem of \$75 is distributed for the entire travel days, covering breakfast (\$15), lunch (\$20), and dinner (\$40).
-0.000	Per diem stipends are allocated to cover food and beverage expenses incurred during out-of-state business assignments.
-0.000	* **Tier 2 Destinations (Other Cities):** Maximum allowable per diem rate is \$150 per night.
-0.000	* **Tier 1 Destinations (Major Metros):** Maximum allowable per diem rate is \$250 per night, exclusive of local lodging taxes.
0.000	When traveling on official business, accommodation expenses are capped based on the destination tier system to ensure fiscal responsibility.

Retrieval failure + hallucination

Context Attribution

Upload test file (.json)

case_05_hallucination.json

3.7 KB ↓

Query

How much does the Lumen Arc cost, and when will it ship?

Run

The Lumen Arc costs \$899 and ships on March 15, 2026. Beyond pricing, the glasses weigh 48 grams, use micro-LED projectors rated for outdoor visibility, and last a full day of typical glanceable use on a single charge. They run ArcOS and pair with both Android and iOS phones.

Select a span to attribute

Full endpoint output

Context attribution

Score	Source
37.622	The Arc weighs 48 grams, about the same as a regular pair of sunglasses, and uses a pair of micro-LED projectors rated at 400 nits for outdoor visibility.
21.386	Lumen claims the Arc runs for a full day of typical glanceable use on a single charge, with a companion carrying case providing two additional recharges.
18.504	The glasses run ArcOS, a stripped-down platform that pairs with both Android and iOS phones, and Lumen has opened a software development kit so outside developers can build their own overlays.
1.212	Lumen Devices today unveiled the Lumen Arc, its first pair of augmented-reality glasses, at the company's annual developer conference in Austin.

Context Attribution

Upload test file (.json)

case_05_hallucination.json

3.7 KB ↓

Query

How much does the Lumen Arc cost, and when will it ship?

Run

Attributing chars [0, 54]:

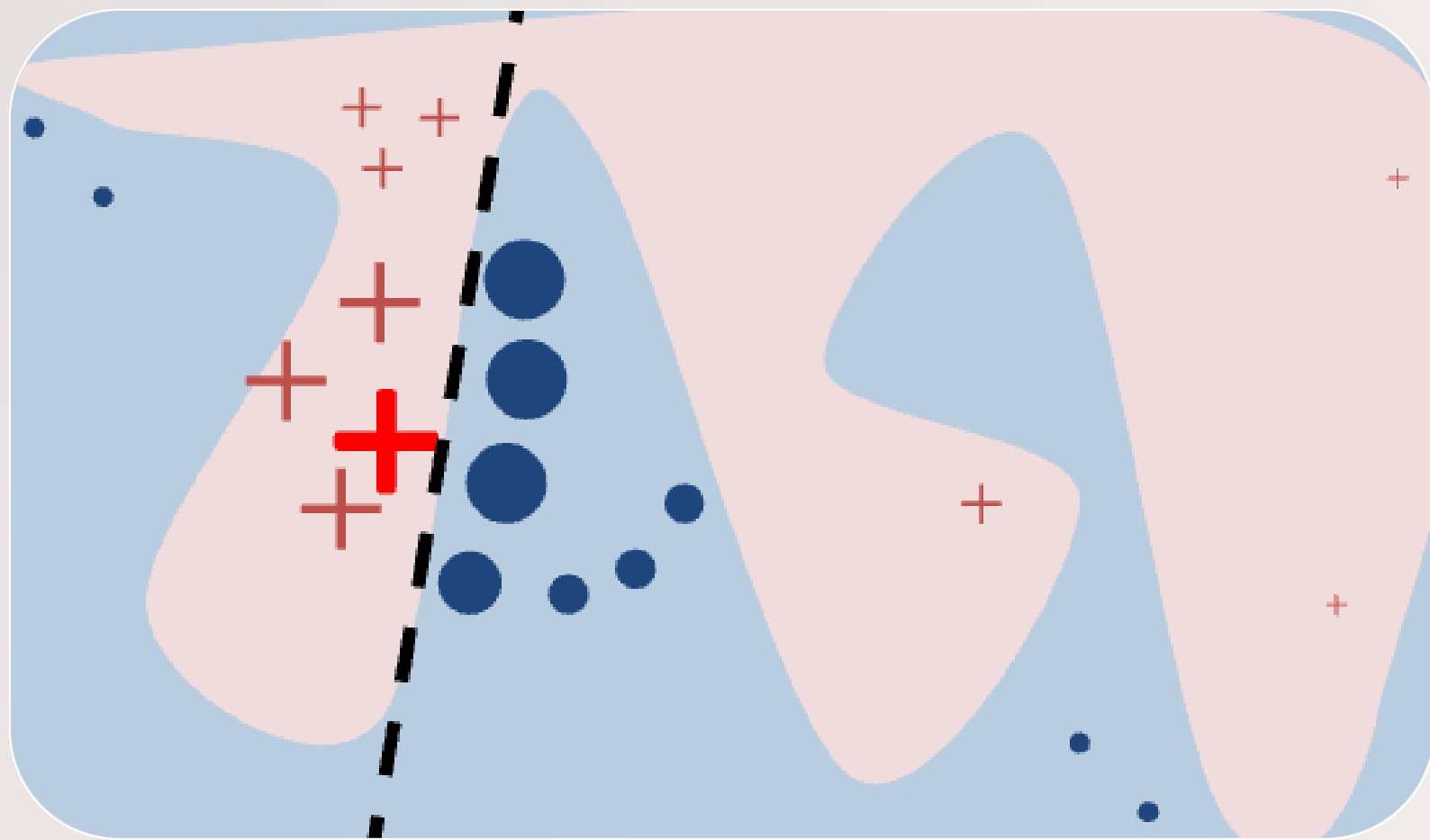
The Lumen Arc costs \$899 and ships on March 15, 2026.

The Lumen Arc costs \$899 and ships on March 15, 2026. Beyond pricing, the glasses weigh 48 grams, use micro-LED projectors rated for outdoor visibility, and last a full day of typical glanceable use on a single charge. They run ArcOS and pair with both Android and iOS phones.

Context attribution

Score	Source
0.390	Executives repeatedly declined to discuss pricing during the event, saying only that the company would share more on cost and timing closer to launch.
0.095	Pre-orders are not yet open.
0.072	Lumen Devices today unveiled the Lumen Arc, its first pair of augmented-reality glasses, at the company's annual developer conference in Austin.
0.000	Okafor said the team deliberately left out a built-in speaker to keep the frames light, relying instead on bone-conduction audio routed through the temples.
-0.000	We didn't want to build a screen you wear on your face, we wanted something you'd forget you had on, said Lumen chief executive officer.

Hallucination

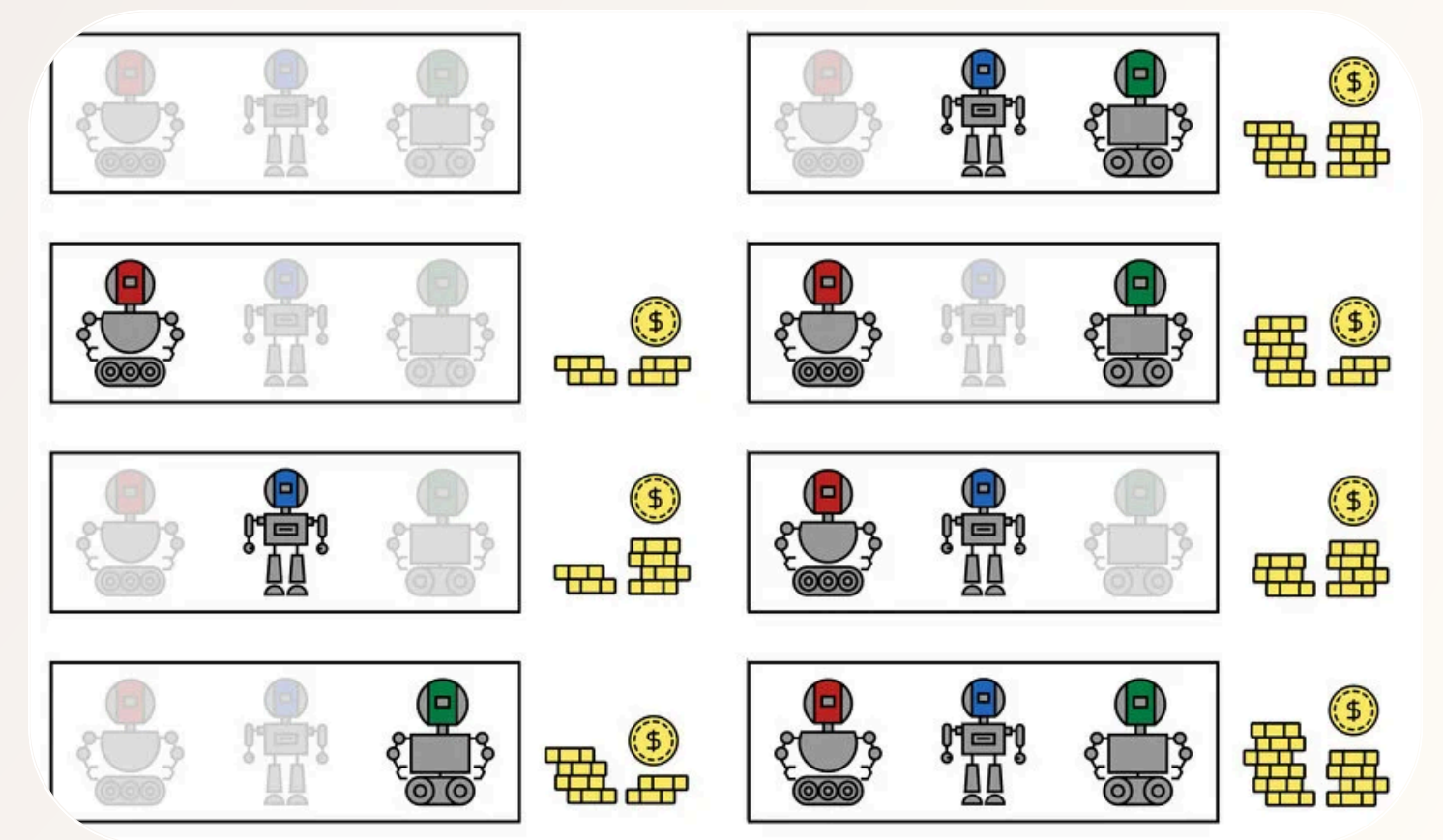


LIME

- Fits perturbed data of an input to simple linear model

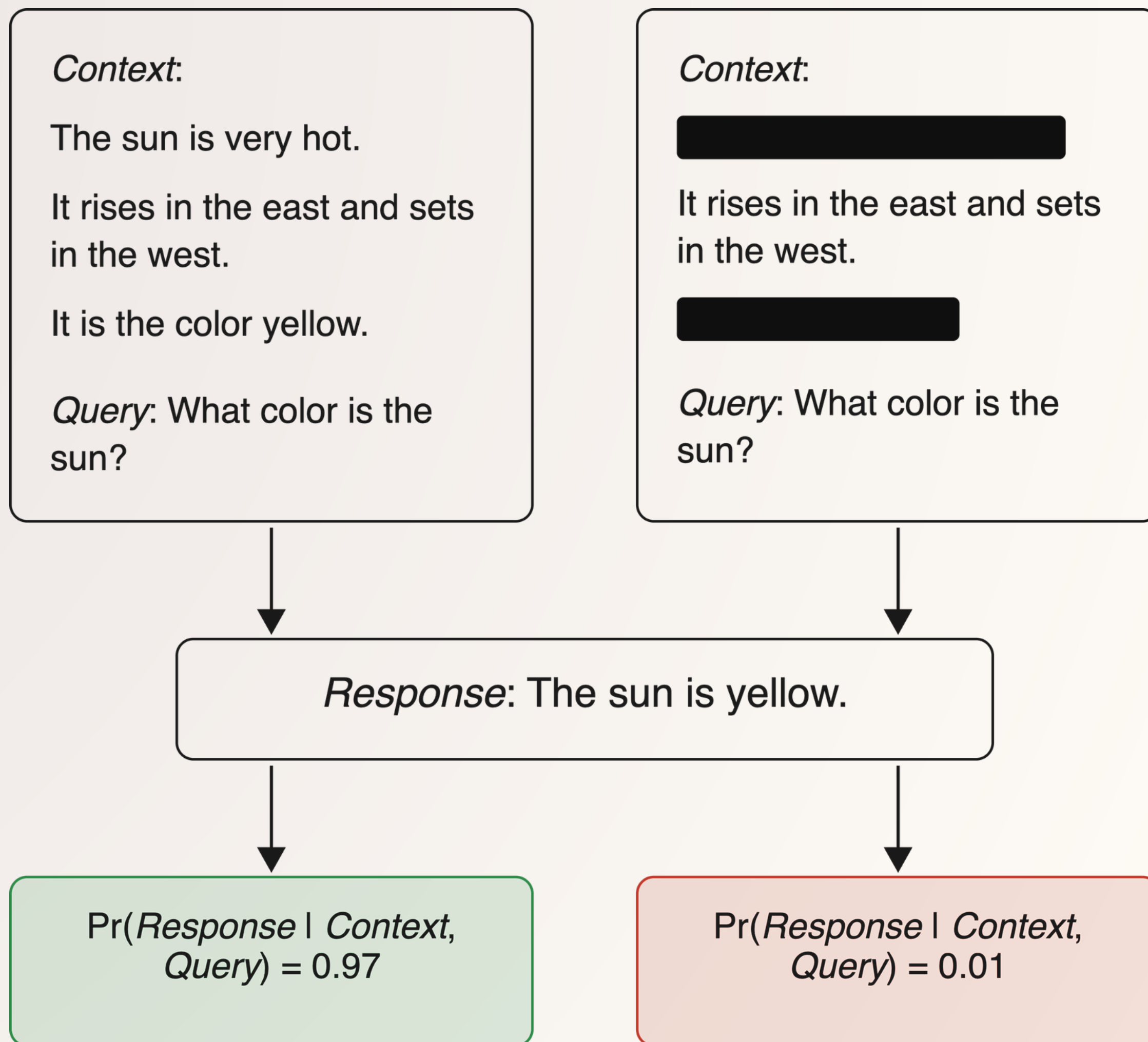
SHAP

- Calculates the contribution of a feature across all feature combinations



Ablation-based attribution is powered by **ContextCite**.

- LLM --> probability of response | context + query
- Probabilities --> fit to a linear model
- Context unit's weight = attribution score

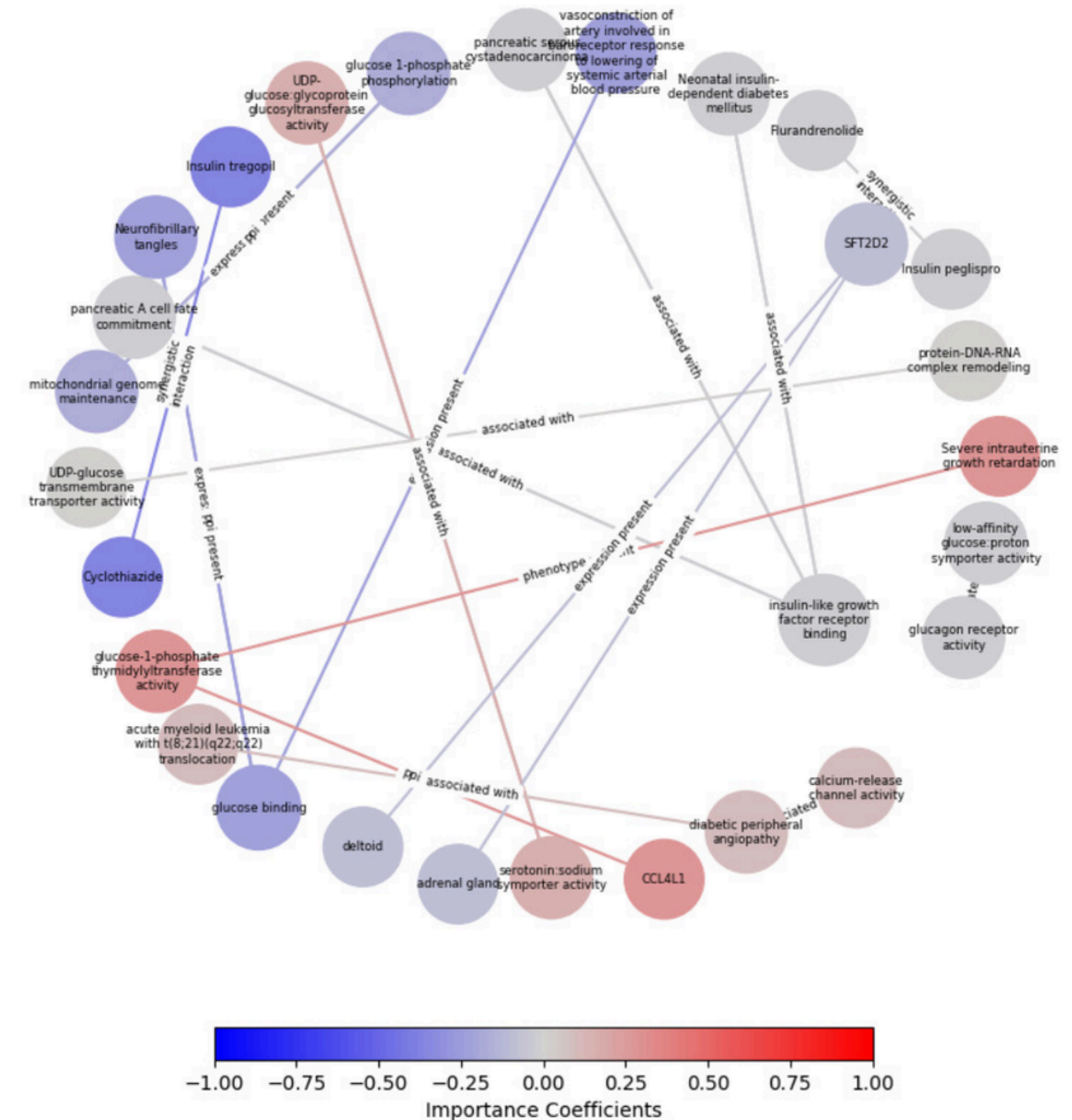


AttriBoT

- LOO but cheaper
- KV caching
- Hierachial attribution
- Proxy modelling

KG-SMILE

- ContextCite for knowledge graph



Future work

Current limitations

- Near duplicate context units
- Low entropy facts (integers)

Next steps

- Instruction ablation
- Automatic failure detection
- KG-SMILE

Thank you